APEX CODERS

Knowledge Handover Platform

Round 1 — Project Plan & Proposal

Prevent Critical Knowledge From Leaving With People

Team Lead	Aayush Patil
Team Member	Omkar Patil
Submission Stage	Round 1 — The Plan
Participant / Institution Details	To be completed if required by the official form

Proposal status: This document is a plan, not a claim that the product has already been built. Technology choices and individual responsibilities below are proposed and should be adjusted to match the team's actual decisions and skills.

1. Understanding the Problem Statement

When an employee, volunteer, or student team member leaves a project, important knowledge can leave with them. The information that remains is often fragmented across documents, code repositories, task boards, meeting notes, and communication. Even when files exist, they may not explain why a decision was made, what is unfinished, which dependencies matter, who owns a task, or what the next person should do.

A generic summarisation tool is not enough. The incoming person needs a trustworthy, practical picture of the project's current state, with important statements linked to supporting evidence. The system must distinguish verified information from uncertainty, missing evidence, and conflicting sources. A responsible person must be able to correct and approve the handover before it is shared.

Core problem we aim to solve: reduce the time and confusion involved in transferring project knowledge by turning authorised project information into an evidence-linked, role-appropriate continuation plan.

Why existing documentation can fail

- Knowledge is scattered across multiple tools and formats.
- Decisions and their reasoning may be buried in conversations.
- Task ownership, blockers, and dependencies can become outdated or unclear.
- A summary without source links can be difficult to verify.
- Sharing all imported information can expose confidential material to people without permission.

2. Users Affected by the Problem

Primary users

- **Outgoing team members:** need a structured way to capture work in progress, decisions, blockers, and context before leaving.
- **Incoming team members:** need to understand the current project state and continue work without repeatedly asking others for basic context.
- **Project leads and managers:** need to validate the handover, confirm ownership, resolve gaps, and approve what is shared.

Other relevant users

- **Student project teams and research groups:** face transitions when members change semesters, teams, or responsibilities.
- **Volunteer and community organisations:** may depend on a small number of people who hold informal operational knowledge.
- **IT and operations teams:** need clear records of dependencies, recurring issues, access constraints, and next actions.

Initial focus: small project teams, including student teams, where project information is spread across documents, tasks, notes, and repositories. This is a proposed starting scope, not a claim that the problem is limited to these users.

3. Proposed Solution

We propose a Knowledge Handover Platform that ingests authorised project sources and builds a reviewable snapshot of the project's current state. Instead of producing one unqualified block of text, it organises information into decisions, completed work, work in progress, unresolved questions, owners, blockers, dependencies, risks, and recommended next actions.

Key differentiator: evidence-first handover

Each important generated claim should include a reference to its source, such as a document section, task, repository item, or meeting note. The interface will mark claims as supported, uncertain, conflicting, or missing evidence where appropriate. Users can inspect sources, edit or reject claims, and approve the final handover.

Proposed handover sections

- Project overview and current status.
- Key decisions, with dates and source references where available.
- Completed work and work in progress.
- Open questions, blockers, dependencies, and risks.
- Task ownership and suggested next actions.
- Known gaps, conflicting evidence, and information requiring confirmation.
- Role-appropriate handover view and approved sharing settings.

The platform will not treat AI-generated assumptions as verified facts. It will only use sources the user is authorised to provide or connect, and the final handover will remain subject to human review.

4. Complete Product Workflow

1. Create project — An authorised user creates a project workspace and chooses the intended incoming role or audience.

2. Connect / import — The user uploads or connects permitted documents, task exports, meeting notes, or repository information. The initial MVP may use file uploads and sample data before adding live integrations.

3. Extract and organise — The system extracts readable content, records source identifiers, and organises evidence by project, date, topic, and source type where possible.

4. Reconstruct project state — AI identifies decisions, ownership, completed and unfinished work, blockers, dependencies, risks, and next actions from the imported evidence.

5. Attach evidence and flag gaps — Important claims link to their supporting source. Unsupported, uncertain, missing, or contradictory information is clearly flagged rather than silently filled in.

6. Review and correct — A project lead or authorised reviewer checks claims, opens sources, edits or rejects inaccurate items, and resolves or acknowledges gaps.

7. Approve and control access — The reviewer approves a version and chooses what the intended recipient is permitted to see. Sensitive content is excluded or redacted where needed.

8. Share and continue — The incoming member receives the approved handover, acknowledges receipt if supported, and uses the prioritised next-action list to continue the project.

MVP boundary

To keep the project feasible, the first working version will prioritise file import, evidence-linked extraction, a structured handover, uncertainty flags, human review, and controlled sharing. Live integrations with every communication and task platform are future scope unless time and authorised access allow them.

5. Main Features We Intend to Build

Feature	Planned behaviour
Authorised source import	Upload project documents and notes; support selected text-based formats and a clearly defined sample dataset.
Project-state reconstruction	Organise current status, decisions, work in progress, blockers, risks, dependencies, ownership, and next actions.
Evidence-linked claims	Show a source reference alongside important claims so reviewers can verify them.
Uncertainty and conflict flags	Mark missing evidence, contradictory details, and items requiring human confirmation.
Human review and approval	Allow authorised reviewers to edit, reject, approve, and version a handover.
Role-appropriate sharing	Generate a focused view for the incoming role and enforce access checks before showing protected content.
Continuation checklist	Provide prioritised next steps, open questions, and unresolved blockers.
Audit-friendly records	Record relevant approval and version events where feasible, without exposing restricted source content.

6. Proposed Technology Stack

The following stack is a practical proposal for an MVP. Final choices should be confirmed against team skills, event rules, available APIs, and the build window.

Layer	Proposed choice and purpose
Frontend	React with Vite; responsive UI for project setup, evidence inspection, review, and handover.
Backend	Python with FastAPI for ingestion, processing, authentication checks, and API endpoints.
AI / language processing	An LLM API for structured extraction and handover drafting, combined with explicit prompts and output validation. Model and provider to be selected by the team.
Document processing	Python libraries appropriate to the chosen formats, such as PDF and DOCX text extraction; unsupported or scanned files should be reported clearly.
Database	PostgreSQL for projects, users, source metadata, claims, review status, versions, and access rules. SQLite may be used for a local prototype if simpler.
Source evidence	Store source identifiers and locations such as file name, page, section, task ID, or repository path wherever available.
Authentication / permissions	Role-based access checks in the backend; do not rely on hiding UI elements alone. Use synthetic or authorised demo data.
Deployment / repository	A web deployment and a version-controlled Git repository with setup instructions; exact hosting provider to be decided.

Security and responsible data handling

- Use synthetic, public, permissioned, or anonymised information for the demo.
- Import only sources the user is authorised to access.

- Apply access checks on the server for every protected project and handover.
- Avoid placing secrets or sensitive source text in logs; protect API keys through environment configuration.
- Make source links and confidence/uncertainty status visible to reviewers.
- Disclose AI APIs, libraries, and reused components as required by event rules.

7. Team Responsibilities

Responsibilities below are a proposed division based on the two team members provided. Confirm or adjust these assignments to reflect actual skills and contributions.

Member	Proposed responsibility	Key deliverables
Aayush Patil Team Lead	Product planning, workflow design, frontend coordination, integration and final submission coordination.	User journey, UI screens, feature prioritisation, integration checks, proposal/demo coordination.
Omkar Patil	Backend and data workflow, source ingestion, AI extraction pipeline, evidence model, testing and documentation.	API/data model, extraction and claim validation, evidence linking, test cases, technical notes.

Both members will contribute to product decisions, end-to-end testing, demo preparation, and explanation of the implementation. These assignments are proposals and must not be presented as completed work until carried out.

8. Development Plan Across Three Rounds

Round 1 — The Plan

- Validate the problem through interviews, observations, or credible supporting research; document what evidence was collected.
- Define the target user, departure scenario, and initial MVP boundary.
- Design the user workflow, data model, evidence strategy, permission model, and interface outline.
- Confirm the stack, task ownership, risks, success measures, and implementation plan.
- Prepare the Round 1 PDF and verify the submission form and shareable PDF access.

Round 2 — The Build

- Build the end-to-end MVP: import authorised sample sources, extract content, generate structured project state, and attach source references.
- Implement uncertainty/conflict flags and a review-edit-approve journey.
- Add a basic role-based sharing path and test that restricted information is not shown to unauthorised users.
- Deploy or prepare a judge-accessible demo, publish the repository as required, and document external APIs and AI assistance.
- Commit to three substantial Round 3 improvements, informed by observed failures and feedback.

Round 3 — The Finish

- Deliver the three improvements committed to in Round 2.
- Improve extraction reliability, evidence quality, handling of missing data, permissions, and error messages.
- Test normal flows and failure cases, including contradictory sources, unsupported files, missing owners, and unauthorised access.
- Prepare a live demo or video, final report, setup instructions, privacy/security explanation, limitations, and roadmap.
- Confirm every team member can explain their contribution and technical decisions.

9. Expected Impact and Success Measures

The intended impact is a faster and more reliable transition when a project member leaves. The platform should help incoming members understand what is known, what remains unresolved, who is responsible, and what action should happen next—without treating unverified AI output as fact.

Measures we plan to evaluate

- **Time to understand:** compare how long a test user needs to answer a fixed set of project-onboarding questions with and without the generated handover.
- **Evidence coverage:** measure the proportion of important claims that have a usable supporting source reference.
- **Issue detection:** check whether missing or contradictory details are flagged in a prepared test dataset.
- **Review quality:** record corrections or rejections made by a reviewer before approval.
- **Permission checks:** test that a user without the required role cannot access a restricted handover or source.

No numerical impact claims are made at this planning stage. Any reported results should come from actual testing and be described with the test method and sample size.

10. Future Scope After the Event

- Optional integrations with authorised task management, source-control, document, and communication platforms.
- More robust support for file types, long projects, multilingual notes, and larger source collections.
- Improved timeline reconstruction and change tracking across handover versions.
- Custom handover templates for engineering, research, operations, student teams, and volunteer organisations.
- Acknowledgement, follow-up questions, and progress tracking for the incoming member.
- More granular permissions, redaction workflows, retention controls, and audit trails.
- Evaluation tools for measuring factuality, citation coverage, and usefulness across different project types.

These are future possibilities, not promised Round 1 or MVP deliverables. Priority will depend on user feedback, technical feasibility, privacy requirements, and event constraints.

11. Team and Participant Details

Field	Details
Team name	Apex Coders
Team lead	Aayush Patil
Team member	Omkar Patil
Number of listed members	2
Institution / college	Vishwakarma Institute of Technology
Course / department	Computer Engineering

End of Round 1 Proposal — Apex Coders